

# Generative Friction as an Alignment Mechanism in Human-AI Ideation

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## Abstract

Seamless generative AI delivers output as a finished product, creating a unidirectional dynamic that constitutes misalignment at the interaction level. We propose Generative Friction, intentional disruptions to AI output (fragmentation, delay, ambiguity), as a mechanism that creates conditions for alignment to emerge through active engagement. In a qualitative study with six designers, friction shifted AI output from product to material, eliciting diverse appropriation strategies. However, responses depended on users' Friction Disposition, their propensity to interpret resistance as invitation rather than obstruction. Generative Friction can work as a complementary, interaction-level approach to alignment mechanism in human-AI ideation.

## CCS Concepts

• **Human-centered computing** → **Interactive systems and tools**; *Natural language interfaces*.

## Keywords

generative AI, friction, ideation, human-AI collaboration, design fixation, appropriation, human-AI alignment, human agency

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## 1 Introduction

Modern generative AI tools are optimised for *seamlessness*—delivering polished, fluent responses instantly. While efficient, this creates a unidirectional dynamic: AI produces, humans consume. We argue this constitutes misalignment at the interaction level. Even when a model is aligned through training-time interventions such as RLHF [1], interaction design can still produce misaligned collaboration if it encourages passive acceptance over active engagement. When a designer copy-pastes AI output verbatim, the resulting artefact reflects AI's patterns rather than the designer's intent—regardless of how well the model was aligned at training time.

Shen et al.'s [2] Bidirectional Human-AI Alignment framework confirms this concern, arguing that alignment has been “approached as a static, one-way process” and that “implicit human signals that

indicate values are frequently overlooked.” We respond by addressing a specific interaction-level layer: *how AI output is presented*, and whether the interface preserves human agency.

Standard alignment framing tends to assume that values precede interaction: the human already knows what they want, and the system's task is to deliver it. Training-time interventions such as RLHF and constitutional AI largely optimise for this logic. Creative work often unfolds differently. Designers may not simply retrieve fully formed values at the start of ideation; rather, they are constituted through friction with material, through resistance that forces choices, through commitments that only become visible in retrospect. In Schön's terms, this is reflection-in-action: knowing emerges through doing s [8]. Seamless AI can disrupt this process, not because it produces incorrect output, but because it produces output that can be accepted without sufficient engagement. What appears to be optimisation with fewer iterations, faster convergence may also optimise away the very activity through which designers discover what they value in the situation. In this sense, seamlessness can produce a form of interaction-level misalignment that training-time interventions cannot detect, because the values at stake are not fully available in advance as a training signal.

In this paper, we propose **Generative Friction**: intentional disruptions that degrade seamlessness to create conditions for active engagement. By fragmenting text (Physical friction), delaying delivery (Temporal friction), or obscuring meaning through metaphor (Semantic friction), we break the illusion of completeness and invite human contribution. While *Protective Friction* [6] adds barriers to verify accuracy—asking “Should I accept this?”—Generative Friction disrupts presentation to ask “What can I *make* from this?” We draw on Dourish's [7] concept of appropriation and Schön's [8] notion of materials that “talk back,” arguing that appropriation behaviours constitute *implicit human feedback* [2], revealing users' creative values through engagement patterns.

## 2 Background

Human-AI alignment is predominantly framed as a training-time challenge [1, 3, 4], but can also break down at the interaction level through automation complacency [9] and design fixation [5]. Cognitive Forcing Functions [6] and Seamful XAI [10] address this in high-stakes domains; we extend it into creative ideation. HCI has increasingly recognised friction's productive potential: Slow Technology reframes delay as reflective time [12]. Seamful Design argues that exposing seams empowers users [13], Cox et al. operationalise microboundaries for mindfulness [14], and Cabitza et al.'s programmed inefficiencies stimulate cognitive engagement [15]. We connect friction to alignment: it is not only a tool for reflection, but a

mechanism that shifts human-AI dynamics from unidirectional consumption to bidirectional collaboration. Critically, appropriation, users adapting technology to their own purposes [7], is a necessary precondition for interaction-level alignment, and prior work supports incomplete AI outputs as generative resources [17, 18].

### 3 Study Design

We conducted a qualitative, within-subjects study with six designers (3F, 3M; ages 22–30; three postgraduate students, three professionals with 2–8 years' experience) using think-aloud protocols. Participants used **SPARK**, a custom AI ideation tool (GPT-4o), under four conditions: Seamless (baseline), Physical Friction (every second word hidden [19]), Temporal Friction (gradual word display [20]), and Semantic Friction (cryptic metaphorical output [17, 21]). Each session comprised pre-task interviews, four 7-minute ideation tasks on student-support design briefs, and post-task interviews. We used reflexive thematic analysis [24], focusing on breakdown-and-repair moments to surface appropriation strategies. Because tolerance for ambiguity [22, 23] was anticipated to influence responses, we assessed each participant's orientation toward AI and ambiguity from pre-task interviews. We acknowledge the limitation that fixed condition order can confound friction type with order and fatigue effects.

### 4 Findings

Friction did not simply “break” workflows; participants actively appropriated disruption, transforming obstacles into creative tools—though this depended on their underlying orientation toward ambiguity.

**Three appropriation strategies emerged.** Under *Physical Friction*, participants shifted to **keyword mining**—scanning fragmented text for actionable terms. P3: “*It was pulling out keywords for me. So I didn’t have to go through the text and highlight.*” Under *Temporal Friction*, participants filled delays with independent ideation through **front-loading**: P5 “*came up with a few ideas while I was waiting... and then you can come back and tweak,*” shifting the dynamic so participants led before consulting AI. Under *Semantic Friction*, metaphorical outputs forced **interpretive labor**: P5 described the experience as working with “a different medium, like jelly,” while P1 treated it as a “mini-game” connecting abstract metaphors to concrete design problems.

**Friction Disposition.** Appreciation and appropriation of friction were not uniform. We identify **Friction Disposition**—a user's propensity to interpret resistance as invitation rather than obstruction—as a preliminary moderating variable, related to tolerance for ambiguity [22] and pre-existing workflow orientation. Three profiles emerged: *Reframers* (P1, P2, P5) interpreted friction as creative constraint—P1: “*I could genuinely feel my head throbbing to think... usually when I’m using AI I’m not thinking.*” The *Native Appropriator* (P3) experienced friction as invisible because their baseline keyword-processing practice already matched the friction affordance, producing the highest idea count under semantic friction. *Resisters* (P4, P6) perceived friction as obstruction—P4: “*I don’t like it at all. It’s just annoying. I want it to give me the ideas so I can move on.*” The P3–P4 contrast most clearly illustrates friction's relational nature: identical conditions produced traction for P3 and drag for

P4, suggesting appropriation depends not on friction design alone but on users' readiness to engage with resistance.

### 5 Discussion and Implications

**The Seamlessness Trap as alignment failure.** P4's seamless-condition behaviour—copy-pasting AI output verbatim—exemplifies interaction-level misalignment: the final output reflects AI's patterns rather than the designer's values. This is not a limitation of model alignment but of interaction design to preserve agency. Shen et al. [2] identify a “significant gap in research on training humans to develop appropriate trust and reliance when collaborating with AI”; the Seamlessness Trap is a concrete instance. Our findings show that a paradox emerges: users who most exhibit passive acceptance are least receptive to friction, echoing Bućinca et al.'s [6] finding that effective interventions receive worst ratings—though our dispositional profiles reveal variance, with Reframers experiencing friction as invitation.

**Friction as precondition for alignment.** Generative Friction does not achieve alignment directly; but it can create conditions under which users can actively engage, generating implicit feedback [2] about their values—the keywords they mine, the ideas they front-load, the interpretive paths they pursue. However, P6's collapse under cumulative semantic friction highlights that friction tolerance may depend on cognitive load, requiring fatigue-aware design. Based on our findings, we propose four design principles for alignment-aware interfaces: *Mode-selectable* (exploration vs. efficiency modes), *Legible* (communicating friction's purpose), *Burden-shared* (progressive disclosure), and *Escapable* (opt-out mechanisms [26]). A system that learns Friction Disposition could adjust friction dynamically, closing the bidirectional loop: humans shape output through appropriation while the system adapts friction in response.

Rather than defaulting to seamlessness, future systems could treat friction as a context-dependent intervention policy: modulating when to be fast and fluent and when to introduce reflective resistance, based on task type, stakes, workflow phase, and user disposition.

### 6 Conclusion

We have argued that alignment is not only a training-time challenge but also an interaction-level one: seamless AI can produce misaligned collaboration even with well-aligned models. Generative Friction can create conditions for active engagement, and the appropriation behaviours it elicits generate implicit feedback about user values—a form of alignment signal that current research largely overlooks [2]. Our findings suggest these mechanisms must account for individual differences, and that users who most need alignment support may be least receptive to it.

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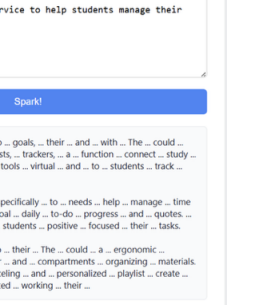
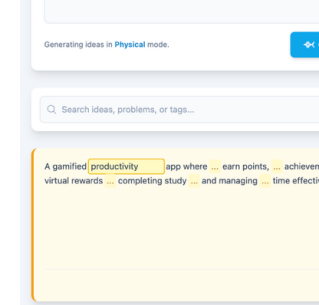
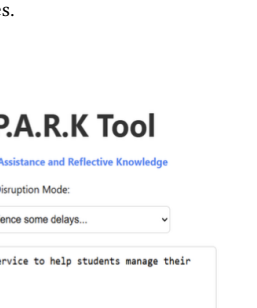
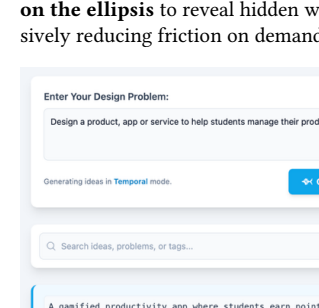
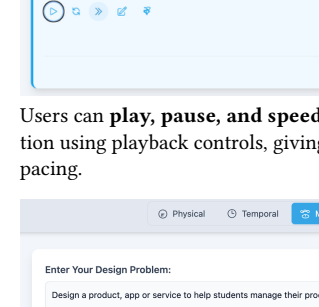
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## A SPARK v1 vs. SPARK v2: Toward Tuneable Friction

Based on our findings, we redesigned the SPARK tool to incorporate tuneable friction controls that allow users to modulate friction intensity—operationalising the design principles proposed in Section 5 and taking a concrete step toward bidirectional alignment. Table 1 summarises the changes.

Table 1: SPARK v1 (fixed friction, used in this study) vs. SPARK v2 (tuneable friction). V2 implements user-controllable mechanisms

| Friction Type | SPARK v1 (Fixed)                                                                    | SPARK v2 (Tuneable)                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|               |    |                                                                 |
| Physical      | <p>Every second word hidden, creating a fragmented display with ellipses.</p>       | <p>Fragmented display retained, but users can <b>click on the ellipsis</b> to reveal hidden words, progressively reducing friction on demand.</p> |
|               |   |                                                                |
| Temporal      | <p>AI output text is displayed gradually over time.</p>                             | <p>Users can <b>play, pause, and speed up</b> presentation using playback controls, giving agency over pacing.</p>                                |
|               |  |                                                               |
| Semantic      | <p>Output presented as cryptic metaphors and riddle-like text.</p>                  | <p>Metaphorical output retained, but users can <b>click on the idea box</b> to reveal a literal description ("explain" toggle).</p>               |